

Load Factor

1. Definition

Formal Definition

Load Factor measures how full a hash table is.

$$\alpha = \frac{n}{m}$$

Where:

- n = number of stored elements
- m = size of hash table
- α = load factor

Interpretation:

- $\alpha = 0 \rightarrow$ table empty
- $\alpha = 0.5 \rightarrow$ half full
- $\alpha = 1 \rightarrow$ completely full
- $\alpha > 1 \rightarrow$ possible only in chaining

2. Intuition Behind Load Factor

Load factor tells us the average number of elements per slot.

Average elements per slot = α

Example:

If:

$$n = 9, \quad m = 11$$

$$\alpha = \frac{9}{11} \approx 0.82$$

Meaning:

On average, each slot is 82% full.

3. Effect on Collisions

Key Observation

As α increases, the probability of collision increases.

When:

$$\alpha \rightarrow 1$$

Almost every insertion causes a collision.

4. Load Factor in Different Techniques

A. Chaining

In chaining:

$$\alpha = \frac{n}{m}$$

Average search time:

$$O(1 + \alpha)$$

- α can be greater than 1
- Table still works
- Performance degrades linearly

B. Open Addressing

In open addressing:

$$\boxed{\alpha < 1}$$

Table must always have at least one empty slot.

As:

$$\alpha \rightarrow 1$$

Performance degrades dramatically.

C. Linear Probing

Expected unsuccessful search:

$$E[U] \approx \frac{1}{2} \left(1 + \frac{1}{(1 - \alpha)^2} \right)$$

When:

$$\alpha = 0.9$$

Performance becomes very poor.

D. Double Hashing

Expected unsuccessful search:

$$E[U] = \frac{1}{1 - \alpha}$$

Performs better than linear probing at high load factors.

5. Visual Understanding

$$\alpha = 0.8$$



80% of the table is filled.

6. When to Resize (Rehashing)

When load factor exceeds a threshold:

$$\alpha > 0.7 \text{ (commonly used threshold)}$$

We:

- Create a new larger table
- Recompute hash values
- Reinsert all elements

This process is called:

Rehashing

7. Practical Recommendations

- For chaining: keep $\alpha \leq 1$
- For open addressing: keep $\alpha \leq 0.7$
- Always use prime table size
- Resize dynamically

Load Factor Controls Performance of Hash Tables